DSA EXPERIMENT 5: Stack and Queue using arrays in C.

NAME:Rushita Kiran Sontakke **PRN:B25ET1051**
BATCH:B1 **CLASS:SY1**

Stack using arrays

```
#include <stdio.h>

#define MAX 5

int stack[MAX], top = -1;

void push(int x)
{
   if (top == MAX - 1)
      printf("Stack Overflow\n");
   else
      stack[++top] = x;
}

void pop()
{
   if (top == -1)
      printf("Stack Underflow\n");
   else
      printf("Element popped: %d\n", stack[top--]);
}

void display()
{
   int i;

   if (top == -1)
      printf("Stack is empty\n");
   else
   {
      for (i = top; i >= 0; i--)
         printf("%d ", stack[i]);
      printf("\n");
   }
}
```

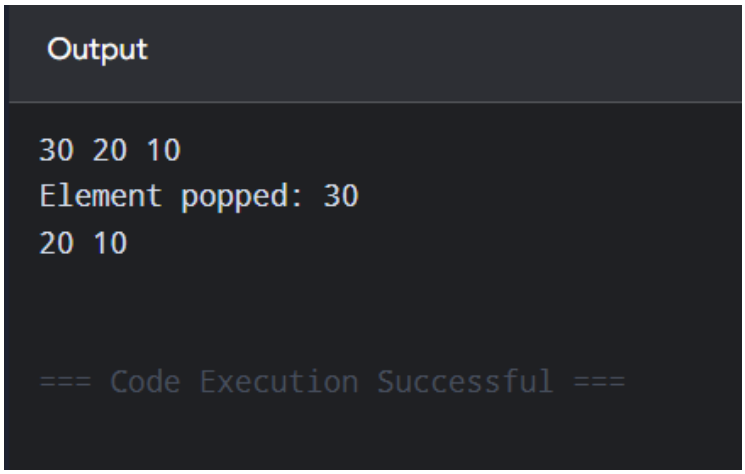
```
int main()
{
    push(10);
    push(20);
    push(30);

    display();

    pop();

    display();

    return 0;
}
```



Queue using array

```
#include <stdio.h>

#define MAX 5

int queue[MAX], front = 0, rear = -1;

void enqueue(int x)
{
    if (rear == MAX - 1)
        printf("Queue Overflow\n");
    else
        queue[++rear] = x;
}

void dequeue()
```

```
{
    if (front > rear)
        printf("Queue Underflow\n");
    else
        printf("Dequeued element: %d\n", queue[front++]);
}

void display()
{
    int i;

    if (front > rear)
        printf("Queue is empty\n");
    else
    {
        for (i = front; i <= rear; i++)
            printf("%d ", queue[i]);
        printf("\n");
    }
}

int main()
{
    enqueue(10);
    enqueue(20);
    enqueue(30);

    display();

    dequeue();

    display();

    return 0;
}
```

Output

```
10 20 30
Dequeued element: 10
20 30


=== Code Execution Successful ===
```